RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Thursday, 8 October 2026, 4:22 PM
State	Finished
Completed on	Thursday, 8 October 2026, 4:24 PM
Time taken	1 min 59 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);
```

```c
    int count = 0;
    int c = 0;

    count++;                            /* i = n/2 initialization */
    for (int i = n / 2; i < n; i++)
    {
        count++;                        /* outer loop condition (true) */
        for (int j = 1; j < n; j = 2 * j)
        {
            count++;                    /* middle loop condition (true) */
            for (int k = 1; k < n; k = k * 2)
            {
                count++;                /* inner loop condition (true) */
                c++;
                count++;                /* c++ */
            }
            count++;                    /* inner loop final (false) check */
        }
        count++;                        /* middle loop final (false) check */
    }
    count++;                            /* outer loop final (false) check */

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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